Answer Key

Quiz 3

Quiz Code: **740190**

Your Performance

15 / 21

71% - Submitted on March 12, 2026 09:23

Quiz Information

Total Questions:	**21**
Time Limit:	**15 minutes**
Quiz Period:	**Mar 12, 2026 09:15 - Mar 12, 2026 14:30**
Answer Key Released:	**March 19, 2026 23:59**

All Questions with Correct Answers & Your Selections

Question 1 You got this right!

1. What is the primary purpose of a Decision Tree in machine learning?

A Clustering data points

B Dimensionality reduction

C **Classification and Regression** Correct Answer Your Answer

D Feature extraction

Question 2 You got this right!

2. Which metric measures the impurity or disorder of a set of items using a logarithmic function?

A Gini Impurity

B Variance

C Mean Squared Error

D **Entropy** Your Answer Correct Answer

Question 3 ✓ You got this right!

3. What does an Entropy of 0 indicate about a dataset?

A It is perfectly balanced between classes

B It contains only missing values

C **It is perfectly pure (all instances belong to one class)**
★ Your Answer – ✓ Correct Answer

D It is completely random

Question 4 ✓ You got this right!

4. How is Information Gain calculated?

A **Entropy of parent - Sum of Entropies of children**
★ Your Answer – ✓ Correct Answer

B Sum of Entropies of children - Entropy of parent

C Gini of parent * Gini of children

D Total Entropy / Total Gini

Question 5

5. What is the maximum possible value of Entropy for a binary classification problem?

A 0.5

B 1 ✔ Correct Answer

C 2

D Infinity

Question 6

6. In a decision tree, what does a leaf node represent?

A A feature to split on

B A missing value

C A test condition

D A class label or final prediction ✔ Correct Answer

Question 7 ✓ You got this right!

7. What describes the "greedy" approach used in standard decision tree induction?

A Evaluating all possible trees

B **Making the locally optimal split at each node** ★ Your Answer ✓ Correct Answer

C Maximizing the depth of the tree

D Using all features at every split

Question 8 ✓ You got this right!

8. The process of dividing data repeatedly based on the feature that maximizes purity is called:

A **Recursive Splitting** ★ Your Answer ✓ Correct Answer

B Cross-validation

C Bootstrap aggregating

D Feature scaling

Question 9 ✔ You got this right!

9. Which of the following is a common stopping criterion for recursive splitting?

A Information Gain reaches 1.0

B All features have been used once

C **A predefined maximum depth is reached** ✔ Correct Answer ★ Your Answer

D Gini impurity is maximized

Question 10

10. What happens if a decision tree is allowed to grow very deep without any restrictions?

A Underfitting

B High Bias

C Low Variance

D **Overfitting** ✔ Correct Answer

Question 11 You got this right!

11. Which of the following is a technique used to combat overfitting in decision trees?

A Increasing maximum depth

B **Pruning** Your Answer Correct Answer

C Adding noisy data

D Reducing the number of training samples

Question 12 You got this right!

12. Overfitted decision trees typically exhibit which characteristics?

A High bias, low variance

B **Low bias, high variance** Your Answer Correct Answer

C High bias, high variance

D Low bias, low variance

Question 13 You got this right!

13. What does "cardinality bias" refer to in the context of decision trees?

A Preferring numerical data over categorical

B **Preferring features with many unique values** Your Answer Correct Answer

C Ignoring the target variable

D Splitting nodes into only two branches

Question 14 You got this right!

14. Which of the following metrics is most susceptible to cardinality bias?

A Gain Ratio

B Gini Index

C **Information Gain** Your Answer Correct Answer

D Variance Reduction

Question 15 You got this right!

15. Why does a feature like "Customer ID" artificially inflate Information Gain?

A **It splits the data into many small, highly pure subsets**
Your Answer – Correct Answer

B It has a very low intrinsic entropy

C It perfectly balances the child nodes

D It merges all classes together

Question 16

16. How does the Gain Ratio metric address cardinality bias?

A It ignores all categorical variables

B **It divides Information Gain by the feature's intrinsic split information**
Correct Answer

C It replaces Entropy with Gini Impurity

D It limits the depth of the tree

Question 17

17. The ID3 algorithm primarily uses which metric to determine the best split?

A Gini Impurity

B **Information Gain** Correct Answer

C Gain Ratio

D Mean Absolute Error

Question 18 **You got this right!**

18. Which splitting criterion does C4.5 use to overcome the flaws of ID3?

A Information Gain

B Gini Impurity

C **Gain Ratio** Your Answer Correct Answer

D Chi-Square

Question 19 ✓ You got this right!

19. A node containing 10 samples of Class A and 10 samples of Class B has a Gini Impurity of:

A 0

B 0.25

C **0.5** ★ Your Answer ✓ Correct Answer

D 1

Question 20

20. If a split results in completely pure child nodes, the Information Gain will be:

A Negative

B Zero

C **Maximized** ✓ Correct Answer

D Equal to the Gini Impurity

Question 21 You got this right!

21. Which of the following best describes the core objective of a splitting criterion?

A **To maximize the homogeneity (purity) of resulting child nodes**

Your Answer – Correct Answer

B To increase the overall depth of the tree

C To ensure all branches have equal numbers of samples

D To use every feature in the dataset